

MUKDAHAN, Thailand

WMO: 483830

Lat: 16.533N

Lon: 104.717E

Elev: 459

StdP: 14.45

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	57.1	59.4	47.5	49.3	63.0	49.8	53.9	64.5	13.8	73.7	11.9	75.6	2.0	50	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.9	100.2	77.5	98.1	77.4	96.1	77.4	81.2	90.9	80.8	90.3	80.3	89.5	3.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	153.5	84.8	78.5	150.5	84.4	78.1	148.4	84.2	45.4	90.7	44.9	90.6	44.4	89.9	91.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	8.6	7.4	DB	53.2	103.2	2.8	1.9	51.2	104.6	49.5	105.7	48.0	106.8	46.0	108.1	
			WB	49.8	83.3	2.6	2.1	47.9	84.8	46.4	86.1	44.9	87.2	43.0	88.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.2	74.4	78.3	83.3	86.3	85.6	84.5	83.0	82.5	82.4	81.3	78.1	74.3
	DBStd	5.57	4.79	6.02	5.63	4.36	3.38	2.79	2.90	2.41	2.37	2.78	3.91	4.75
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	5	2	1	0	0	0	0	0	0	0	0	0	2
	CDD50	11382	757	794	1033	1090	1103	1036	1022	1008	972	970	843	754
	CDD65	5913	294	375	569	640	638	586	557	543	522	505	393	291
	CDH74	64014	2589	3853	6826	8141	7801	6948	6034	5678	5357	4982	3466	2339
	CDH80	27974	1014	1850	3632	4438	3794	3030	2260	1950	1908	1939	1331	828
(14) Wind	WSAvg	1.9	2.4	2.3	2.2	1.7	1.3	1.4	1.4	1.3	1.2	1.9	2.5	2.8
(15) Precipitation	PrecAvg	61.6	0.1	0.7	1.3	3.3	8.0	11.1	10.2	13.4	10.7	2.7	0.2	0.1
	PrecMax	82.5	1.6	2.9	2.9	8.5	13.3	23.1	16.0	24.1	20.2	9.2	1.0	0.7
	PrecMin	45.6	0.0	0.0	0.0	0.7	4.0	5.4	2.5	7.7	3.7	0.6	0.0	0.0
	PrecStd	10.2	0.4	1.0	1.0	1.8	2.8	4.3	3.7	5.1	5.0	2.2	0.3	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.5	99.0	102.1	103.6	101.3	97.2	96.0	93.6	93.5	93.2	94.3	92.6
		MCWB	73.8	74.2	75.9	77.5	78.9	79.7	79.2	80.4	80.1	78.9	77.6	75.6
	2%	DB	91.2	96.0	99.4	100.8	98.0	94.7	92.9	91.8	91.7	91.6	91.9	90.1
		MCWB	72.9	74.2	76.1	78.1	79.1	79.7	79.5	80.0	79.3	78.0	76.7	74.1
	5%	DB	88.7	93.4	97.2	98.4	95.5	92.9	91.0	90.0	90.3	90.3	89.6	87.7
		MCWB	72.1	74.0	76.0	78.3	79.0	79.6	79.4	79.6	78.9	77.3	75.9	72.8
	10%	DB	86.1	90.6	94.8	95.9	93.1	91.1	89.2	88.3	88.7	88.8	87.2	84.9
		MCWB	71.0	73.4	75.9	78.1	78.9	79.5	79.1	79.2	78.6	76.7	74.6	71.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.7	77.9	79.9	81.5	81.8	81.8	81.4	81.6	81.5	80.6	79.4	77.2
		MCDB	88.7	92.6	94.7	96.0	92.7	91.4	89.7	89.9	90.1	90.3	90.3	89.7
	2%	WB	74.9	76.7	78.8	80.3	80.9	80.9	80.6	80.8	80.6	79.5	78.2	75.5
		MCDB	86.5	91.1	93.7	94.5	91.6	90.2	89.1	88.8	88.9	88.6	88.3	87.3
	5%	WB	73.4	75.6	77.8	79.6	80.2	80.5	80.0	80.1	79.9	78.7	77.1	74.1
		MCDB	86.0	88.8	92.1	93.4	90.5	89.4	87.9	87.5	87.5	87.3	86.7	85.2
	10%	WB	72.0	74.5	76.9	78.9	79.7	79.9	79.4	79.6	79.3	77.9	75.8	72.6
		MCDB	83.7	86.8	90.6	92.3	89.7	88.4	86.9	86.4	86.1	85.8	84.8	82.9
(35) Mean Daily Temperature Range	5% DB	MDBR	22.3	22.3	20.3	17.9	14.8	12.4	11.2	10.9	11.9	14.5	17.4	19.5
		MCDBR	24.6	23.9	22.0	20.0	17.7	14.7	14.0	13.6	14.4	16.2	19.5	21.9
	5% WB	MCWBR	8.7	7.1	6.0	5.3	4.6	4.1	4.3	4.6	4.8	5.5	6.9	8.1
		MCDWR	22.3	21.2	20.0	18.1	15.2	13.3	12.7	12.6	12.8	14.5	17.5	20.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.459	0.570	0.659	0.621	0.510	0.474	0.464	0.469	0.504	0.512	0.459	0.456
		taud	2.154	1.859	1.694	1.821	2.149	2.273	2.300	2.274	2.165	2.113	2.238	2.187
	Ebn at Noon	Ebn at Noon	262	237	221	231	254	261	264	265	256	250	260	258
		Edh at Noon	46	65	79	69	49	43	42	43	49	49	42	44
(44) All-Sky Solar Radiation	RadAvg	1578	1667	1651	1749	1694	1570	1444	1404	1452	1578	1537	1491	(44)
(45) RadStd	RadStd	82	102	58	94	95	107	148	102	133	126	59	54	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	+0.65	N/A	N/A	N/A	N/A	+0.25	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.49	N/A	N/A	N/A	+0.29	+0.27	N/A	N/A	N/A	N/A

Nomenclature: See separate page